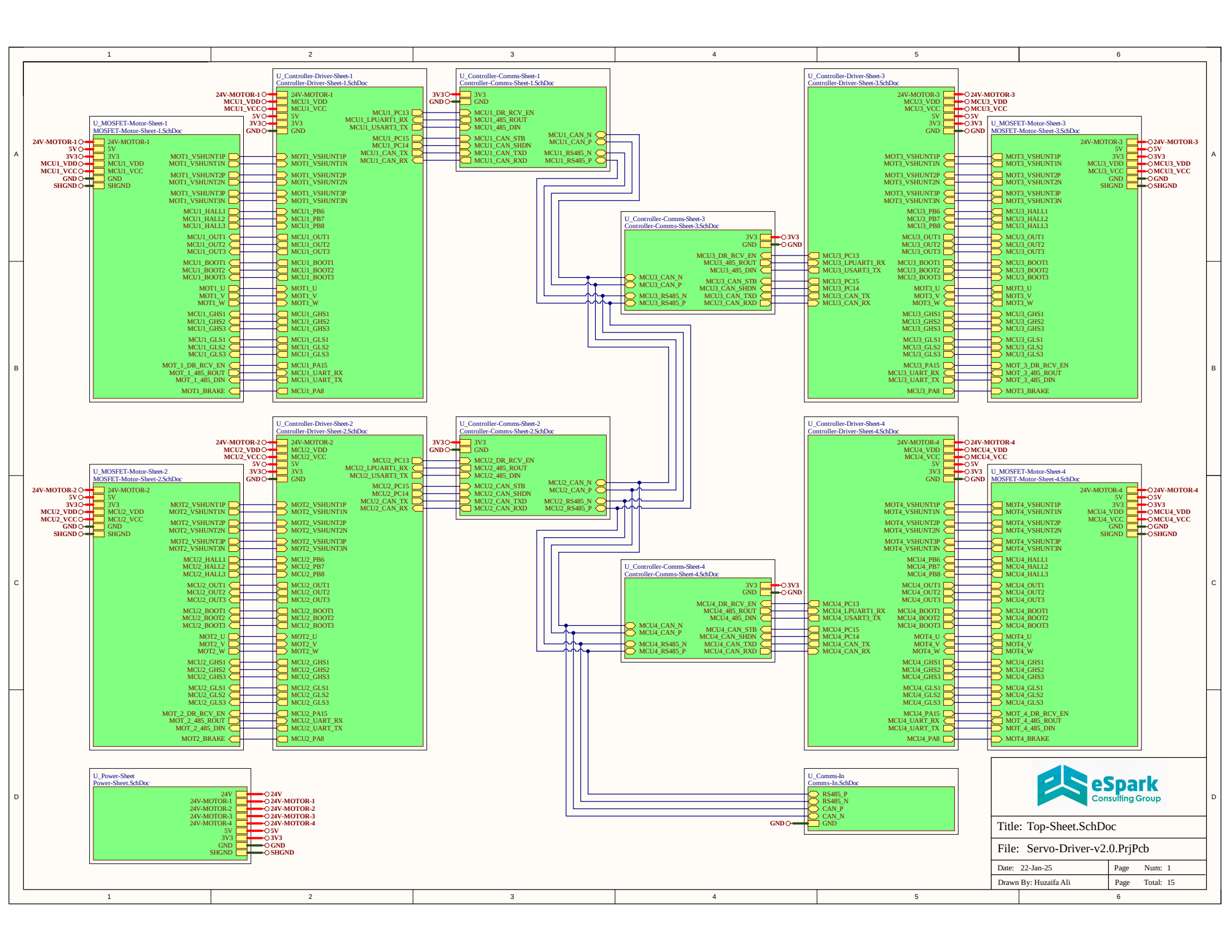
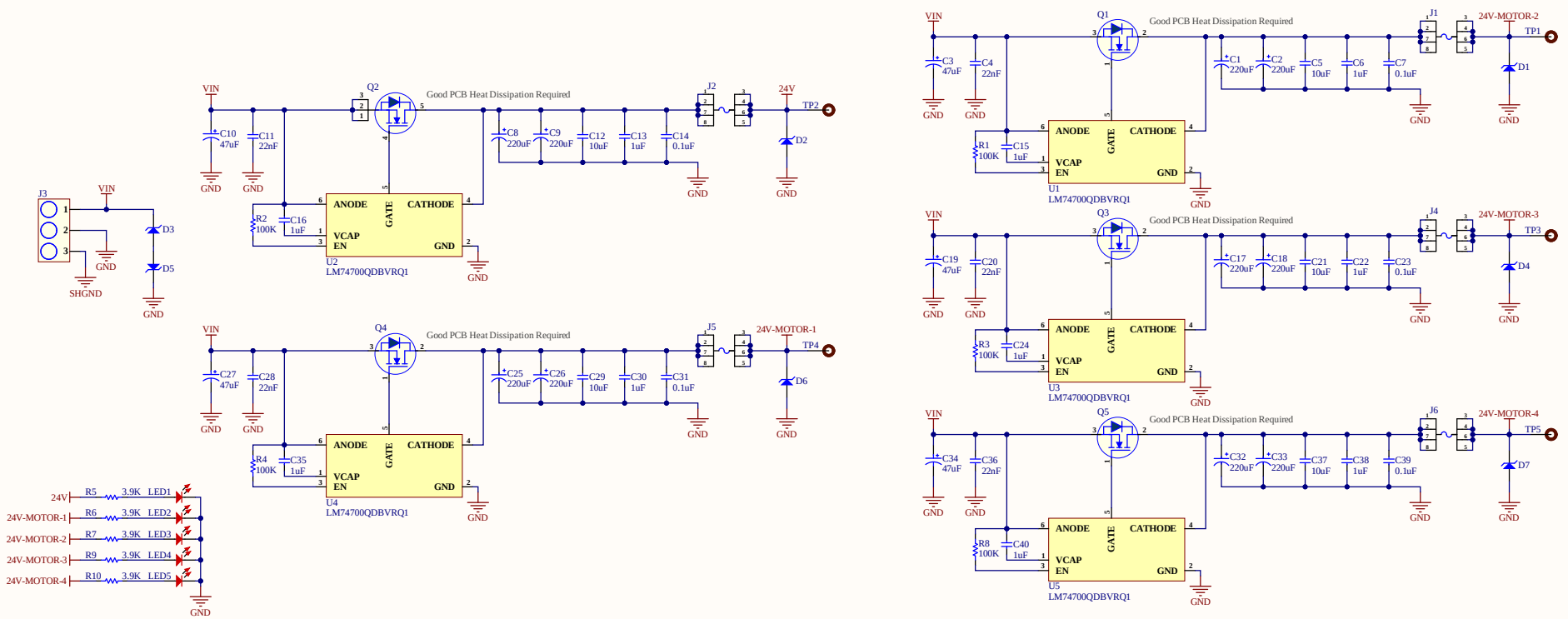
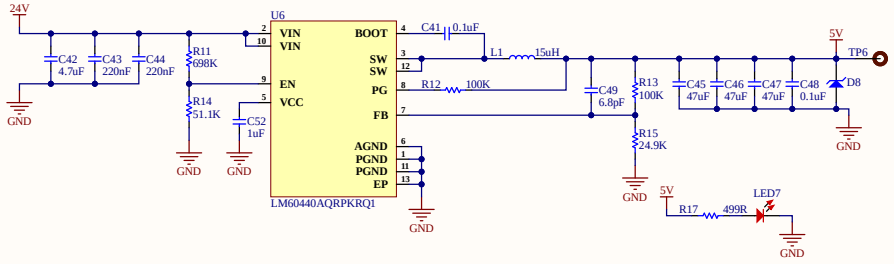

U_MOSFET-Motor-Sheet-1
MOSFET-Motor-Sheet-1.SchDoc
U_Controller-Driver-Sheet-1
Controller-Driver-Sheet-1.SchDoc
U_Controller-Comms-Sheet-1
Controller-Comms-Sheet-1.SchDoc
U_Controller-Comms-Sheet-3
Controller-Comms-Sheet-3.SchDoc
U_Controller-Driver-Sheet-3
Controller-Driver-Sheet-3.SchDoc
U_MOSFET-Motor-Sheet-3
MOSFET-Motor-Sheet-3.SchDoc
U_MOSFET-Motor-Sheet-2
MOSFET-Motor-Sheet-2.SchDoc
U_Controller-Driver-Sheet-2
Controller-Driver-Sheet-2.SchDoc
U_Controller-Comms-Sheet-2
Controller-Comms-Sheet-2.SchDoc
U_Controller-Comms-Sheet-4
Controller-Comms-Sheet-4.SchDoc
U_Controller-Driver-Sheet-4
Controller-Driver-Sheet-4.SchDoc
U_MOSFET-Motor-Sheet-4
MOSFET-Motor-Sheet-4.SchDoc
U_Power-Sheet
Power-Sheet.SchDoc
U_Comms-In
Comms-In.SchDoc
eSpark Consulting Group
Title: Top-Sheet.SchDoc
File: Servo-Driver-v2.0.PrjPcb
Date: 22-Jan-25
Page Num: 1
Drawn By: Huzaifa Ali
Page Total: 15

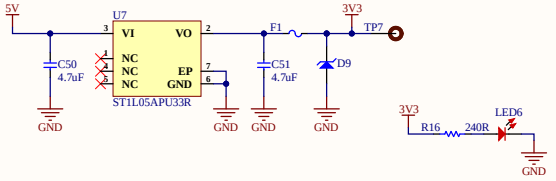
24V INPUT

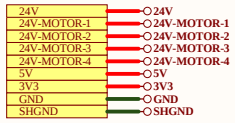


5V BUCK CONVERTER



3.3V LINEAR REGULATOR

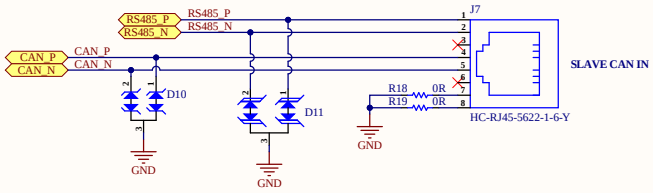




eSpark Consulting Group

Title: Power-Sheet.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 22-Jan-25	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 15

COMMUNICATION IN

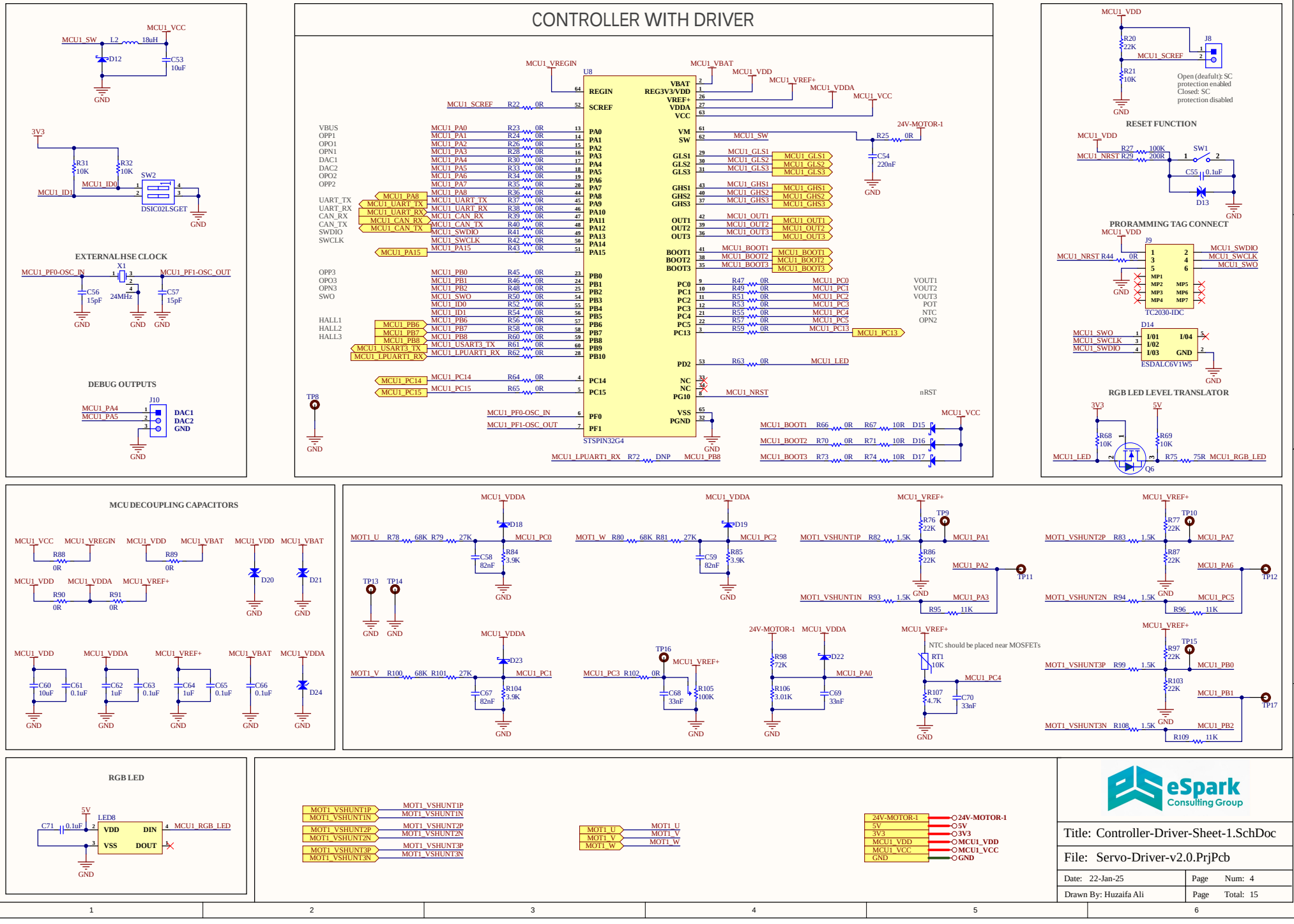


GND — GND

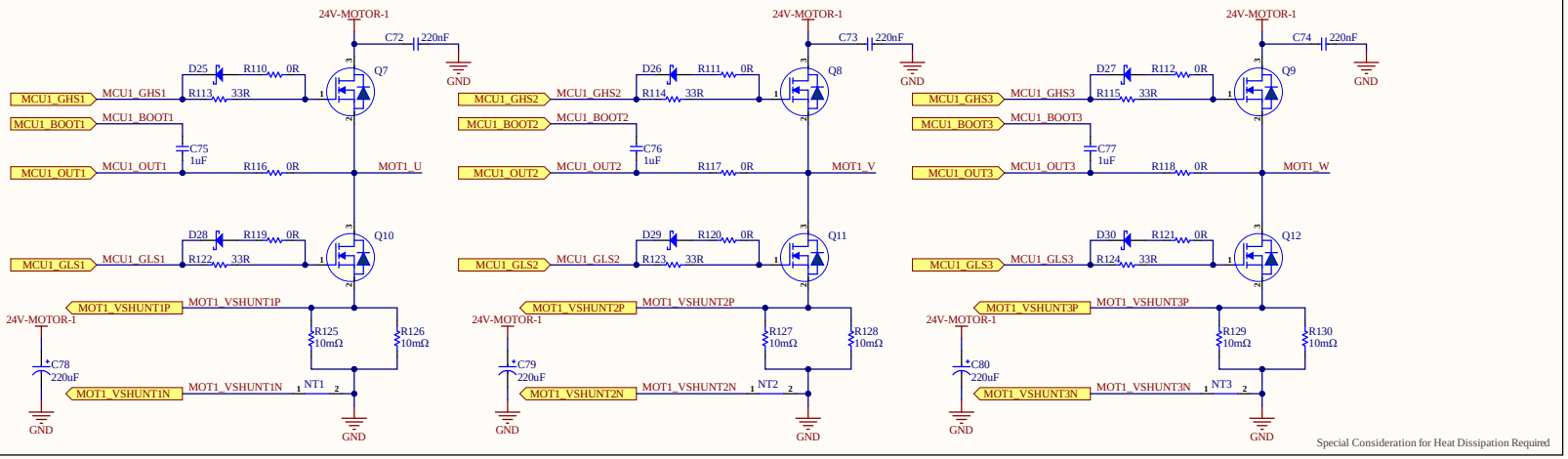
eSpark Consulting Group

Title: Comms-In.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 22-Jan-25	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 15

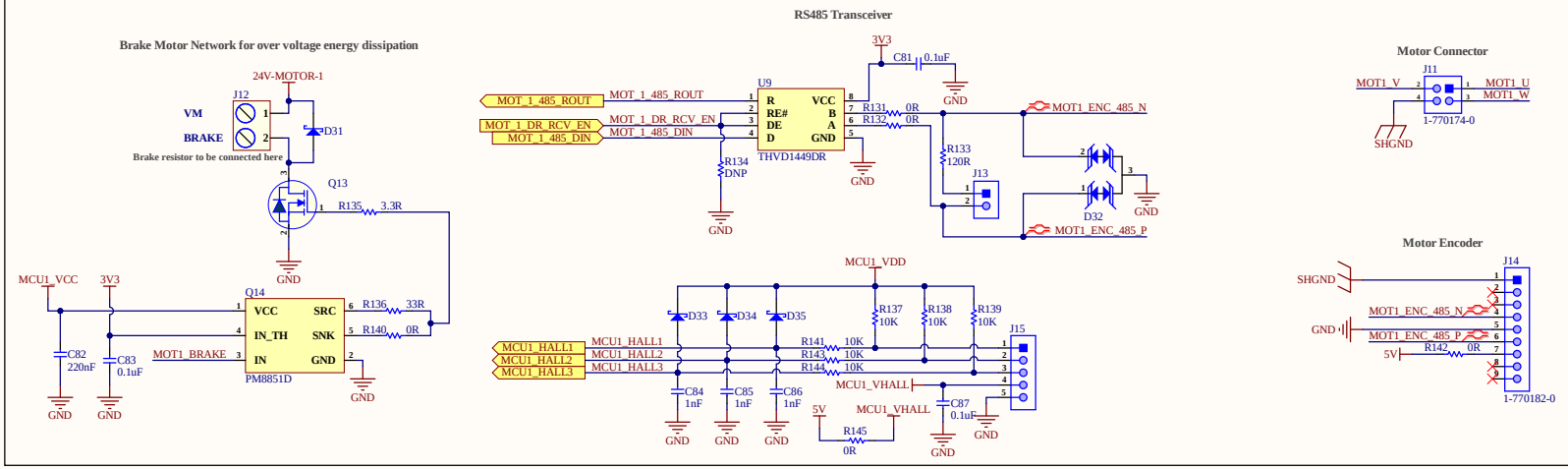
CONTROLLER WITH DRIVER

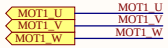


MOSFETs

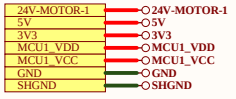


ENCODER AND OTHERS





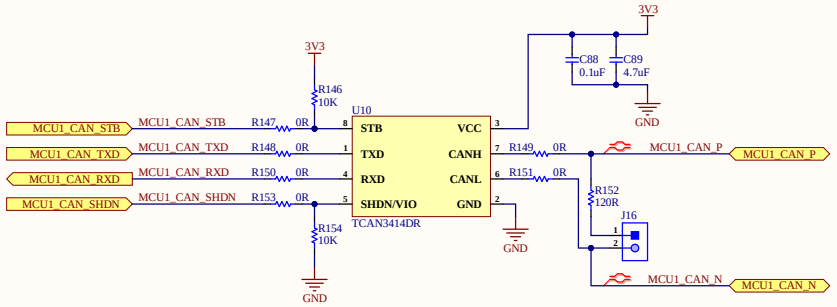




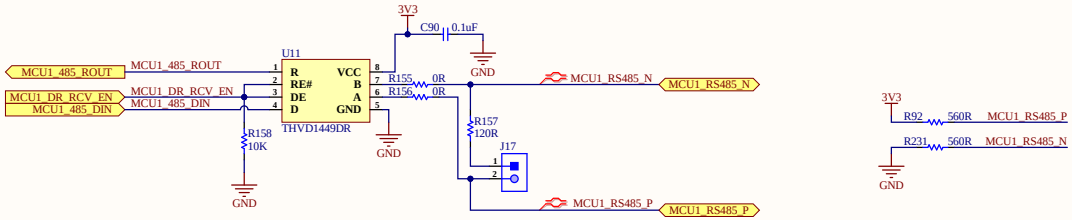
eSpark Consulting Group

Title: MOSFET-Motor-Sheet-1.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 22-Jan-25	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 15

CAN



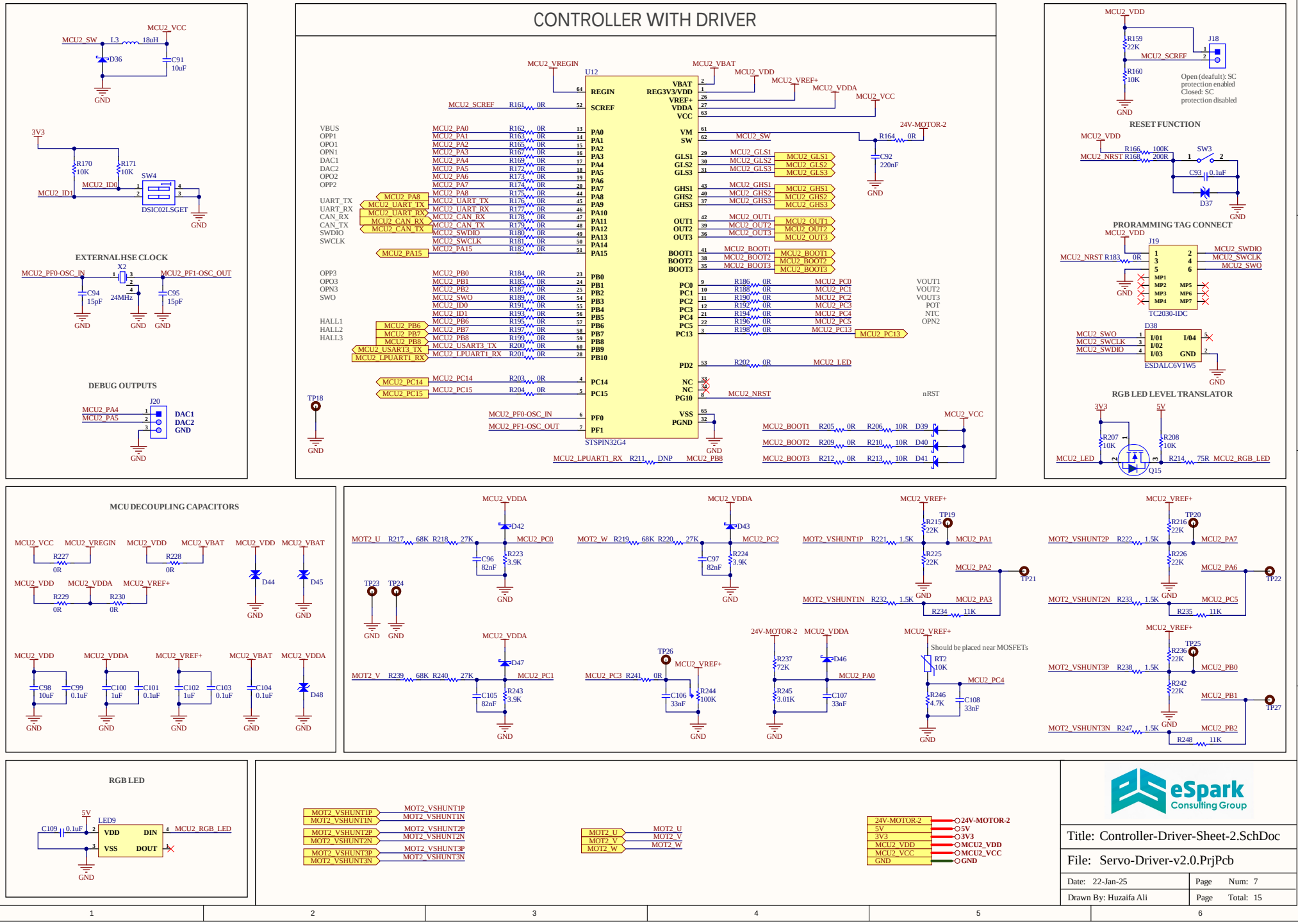
RS485





Title: Controller-Comms-Sheet-1.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 29-Jan-25	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 15

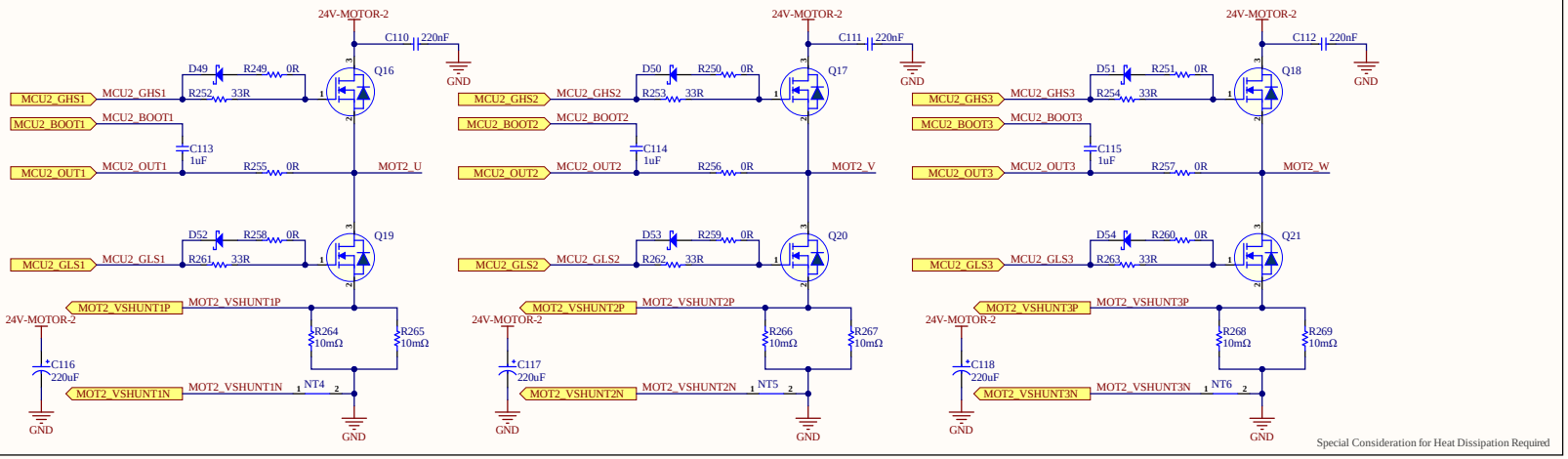
CONTROLLER WITH DRIVER



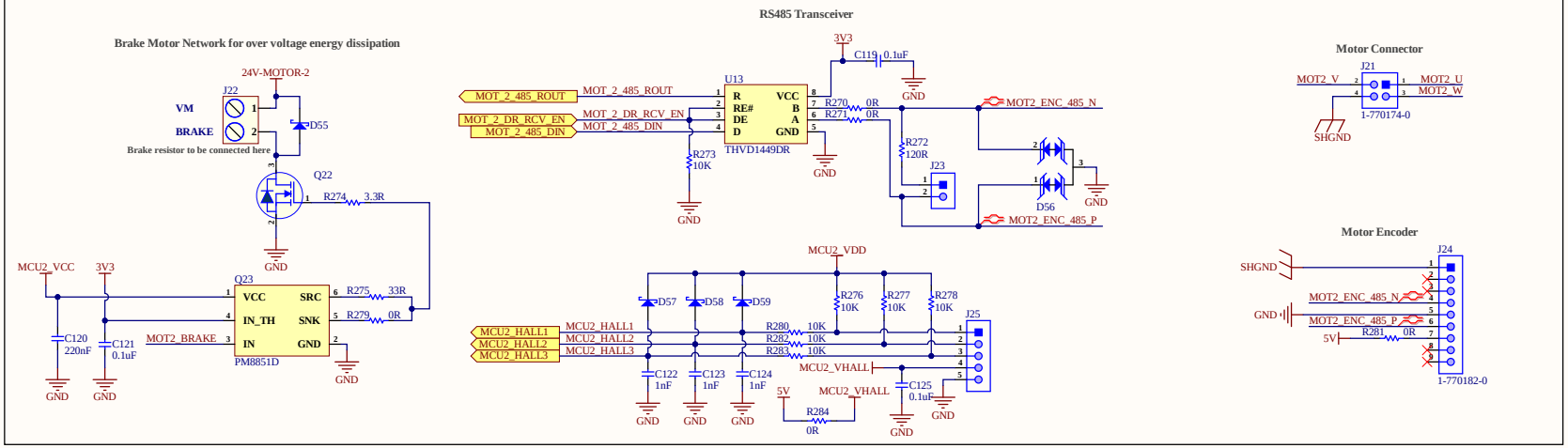
eSpark Consulting Group

Title:	Controller-Driver-Sheet-2.SchDoc
File:	Servo-Driver-v2.0.PrjPcb
Date: 22-Jan-25	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 15

MOSFETs



ENCODER AND OTHERS



MOT2_U
MOT2_V
MOT2_W
MOT2_U
MOT2_V
MOT2_W

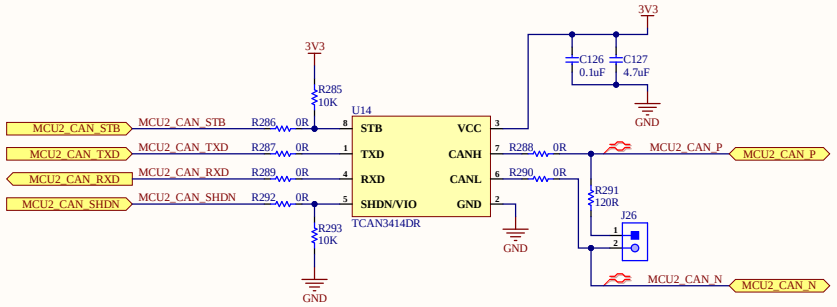
MOT2_BRAKE
MOT2_BRAKE

24V-MOTOR-2
5V
3V3
MCU2_VDD
MCU2_VCC
GND
SHGND
24V-MOTOR-2
5V
3V3
MCU2_VDD
MCU2_VCC
GND
SHGND

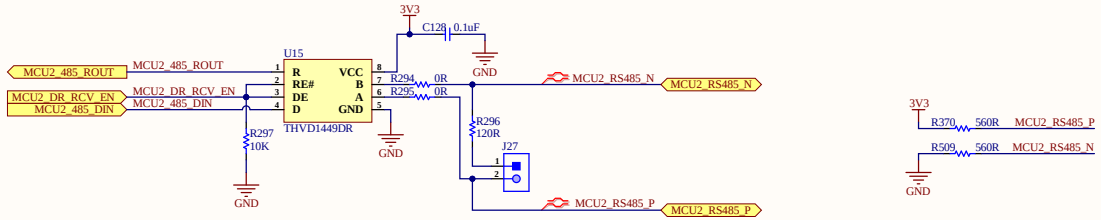
eSpark Consulting Group

Title: MOSFET-Motor-Sheet-2.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 22-Jan-25	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 15

CAN



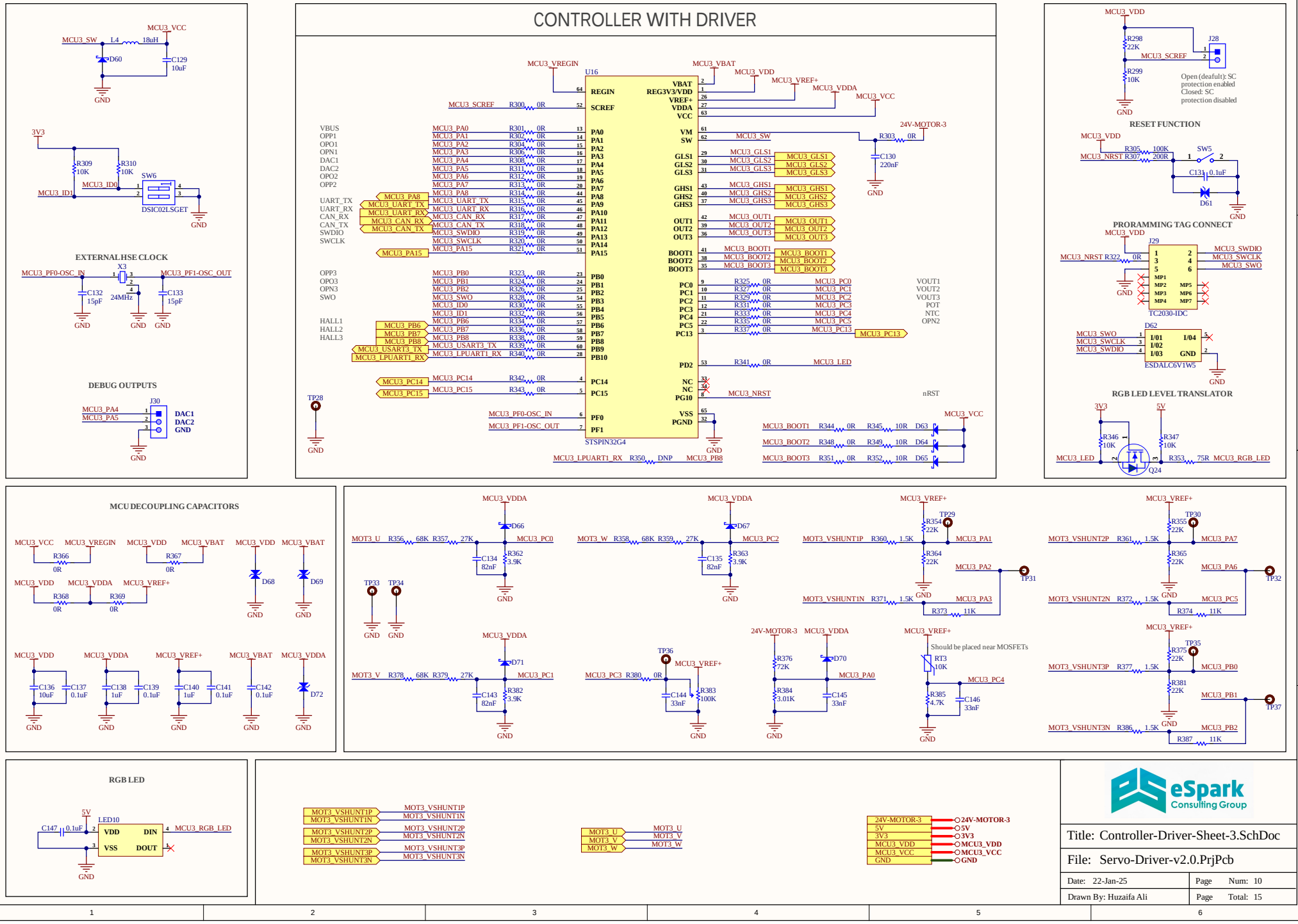
RS485





Title: Controller-Comms-Sheet-2.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 29-Jan-25	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 15

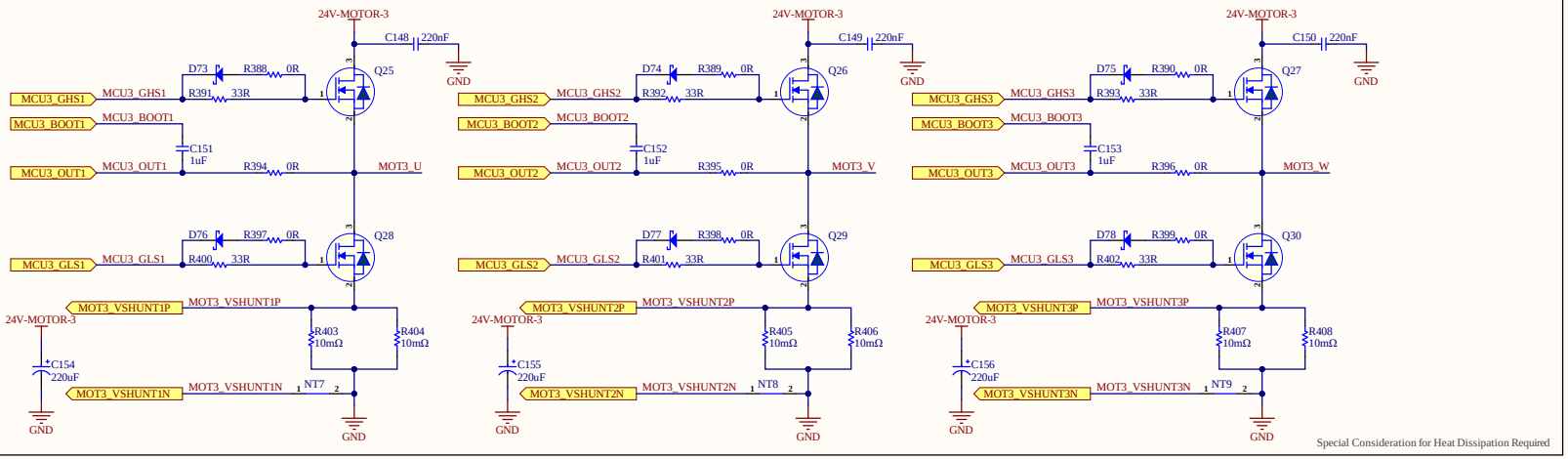
CONTROLLER WITH DRIVER



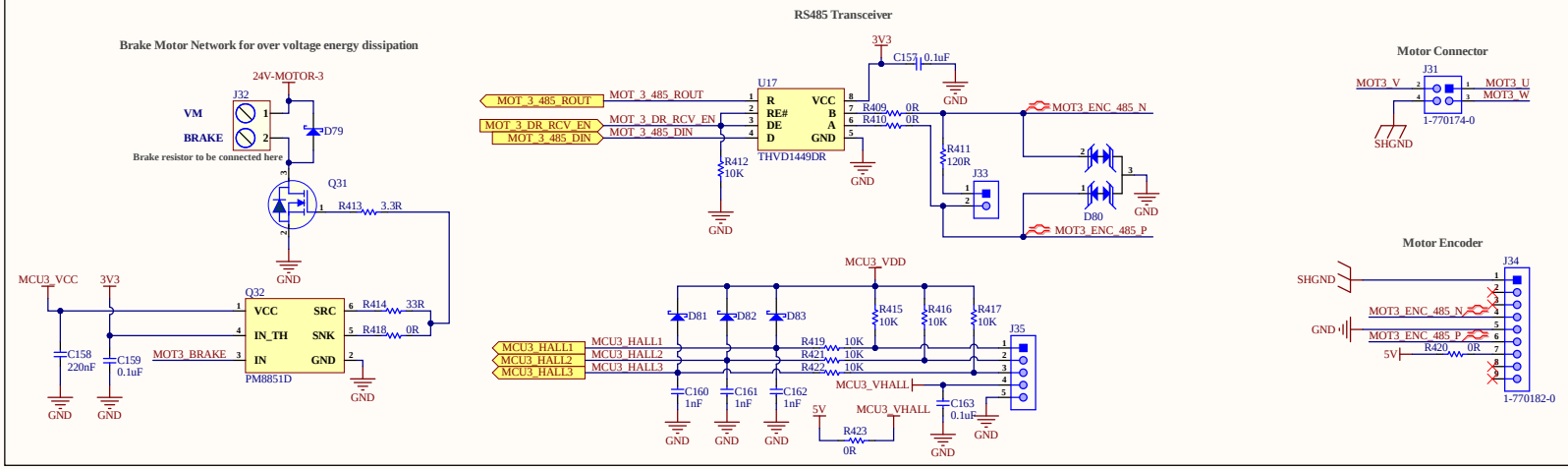
Title:	Controller-Driver-Sheet-3.SchDoc
File:	Servo-Driver-v2.0.PrjPcb
Date: 22-Jan-25	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 15

eSpark Consulting Group

MOSFETs



ENCODER AND OTHERS



MOT3_U
MOT3_V
MOT3_W

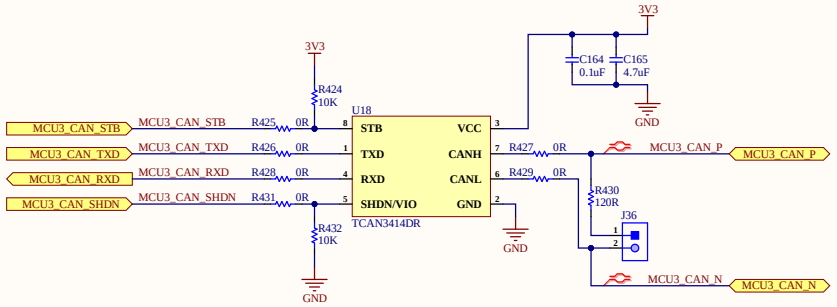
MOT3_BRAKE

24V-MOTOR-3
5V
3V3
MCU3_VDD
MCU3_VCC
GND
SHGND

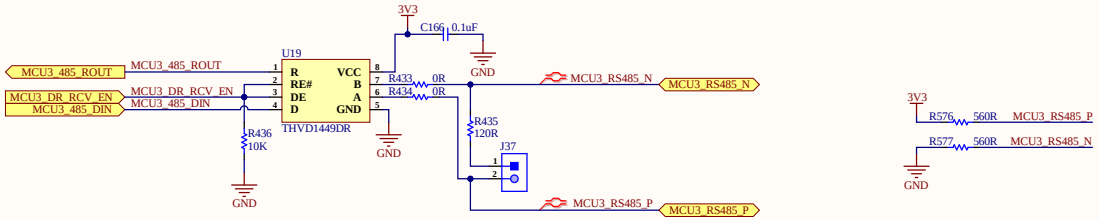
eSpark Consulting Group

Title: MOSFET-Motor-Sheet-3.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 22-Jan-25	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 15

CAN



RS485

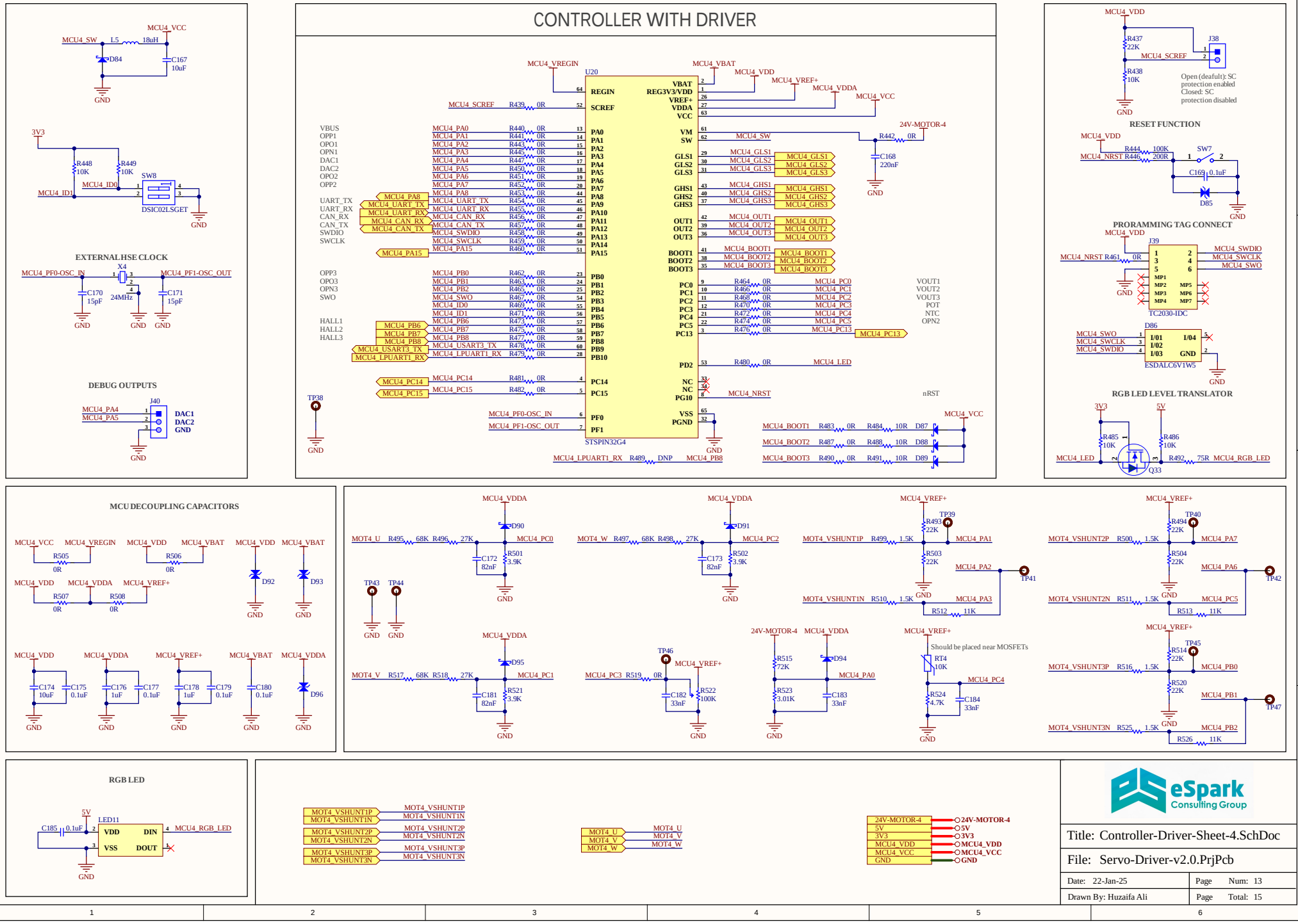


3V3 — 3V3
GND — GND

eSpark Consulting Group

Title: Controller-Comms-Sheet-3.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 29-Jan-25	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 15

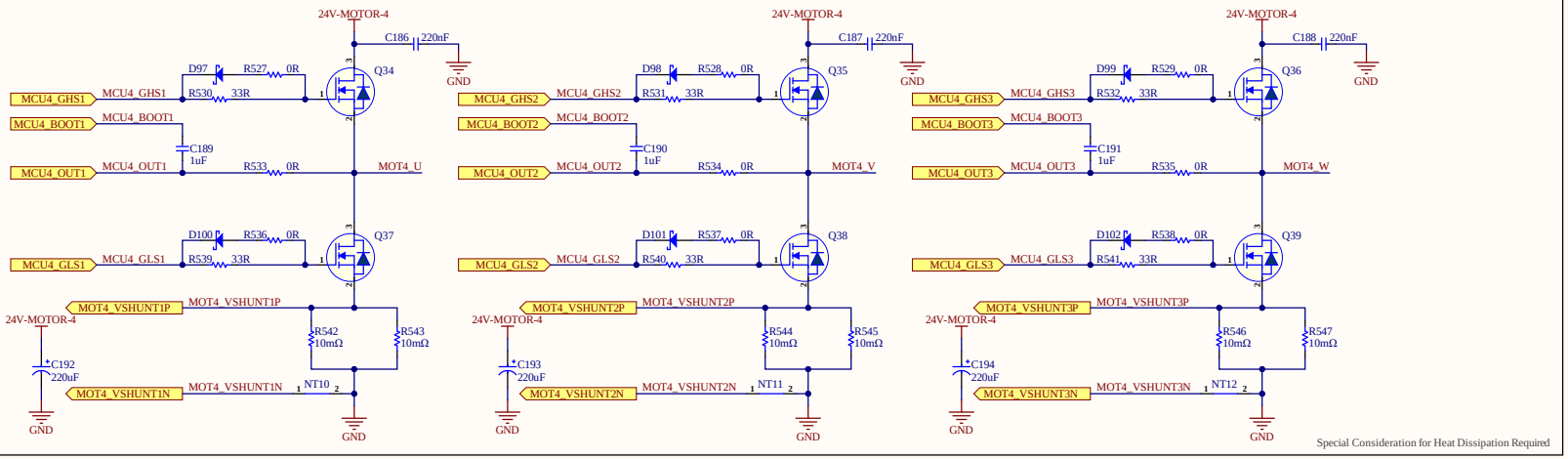
CONTROLLER WITH DRIVER



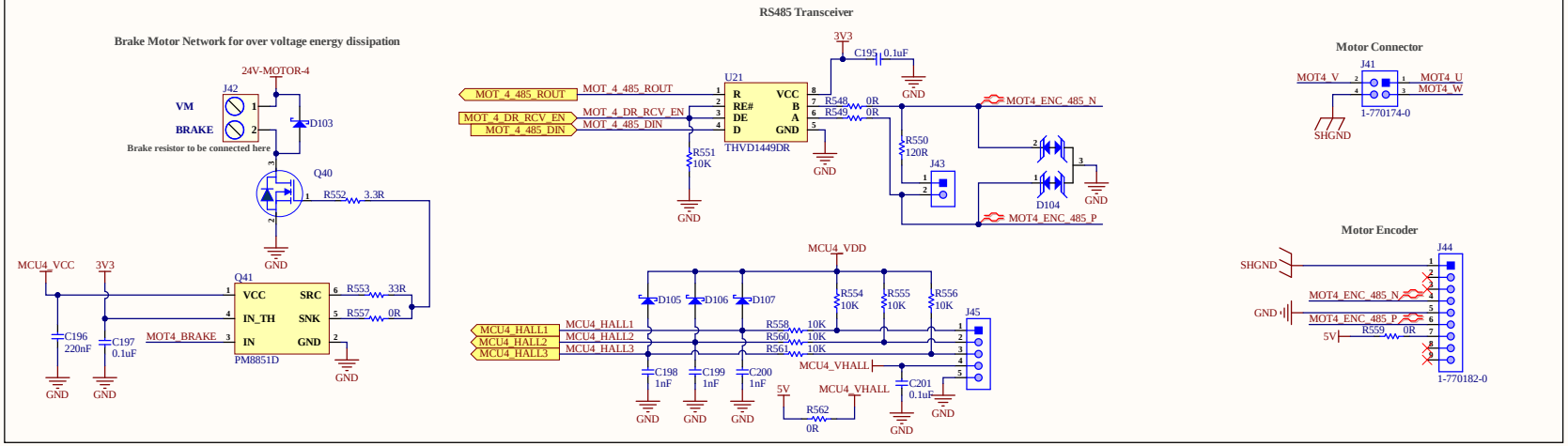
eSpark Consulting Group

Title: Controller-Driver-Sheet-4.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 22-Jan-25	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 15

MOSFETs



ENCODER AND OTHERS



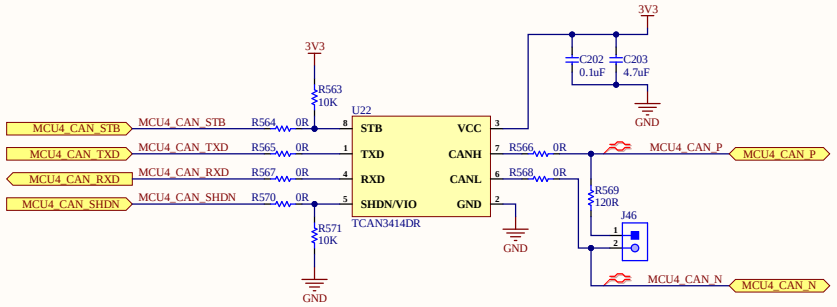
MOT4_U
MOT4_V
MOT4_W

MOT4_BRAKE

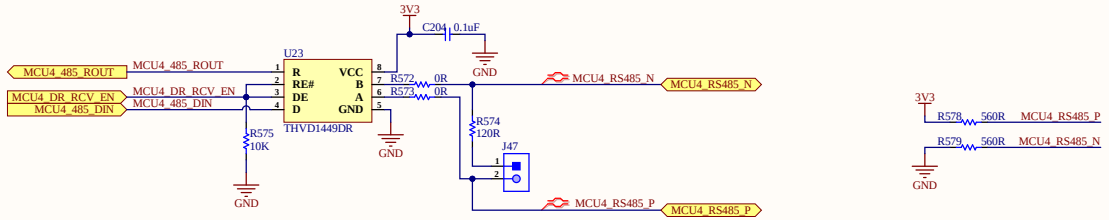
24V-MOTOR-4
5V
3V3
MCU4_VDD
MCU4_VCC
GND
SHGND

Title:	MOSFET-Motor-Sheet-4.SchDoc
File:	Servo-Driver-v2.0.PrjPcb
Date: 22-Jan-25	Page Num: 14
Drawn By: Huzaifa Ali	Page Total: 15

CAN



RS485



3V3 — 3V3
GND — GND

eSpark Consulting Group

Title: Controller-Comms-Sheet-4.SchDoc	
File: Servo-Driver-v2.0.PrjPcb	
Date: 29-Jan-25	Page Num: 15
Drawn By: Huzaifa Ali	Page Total: 15